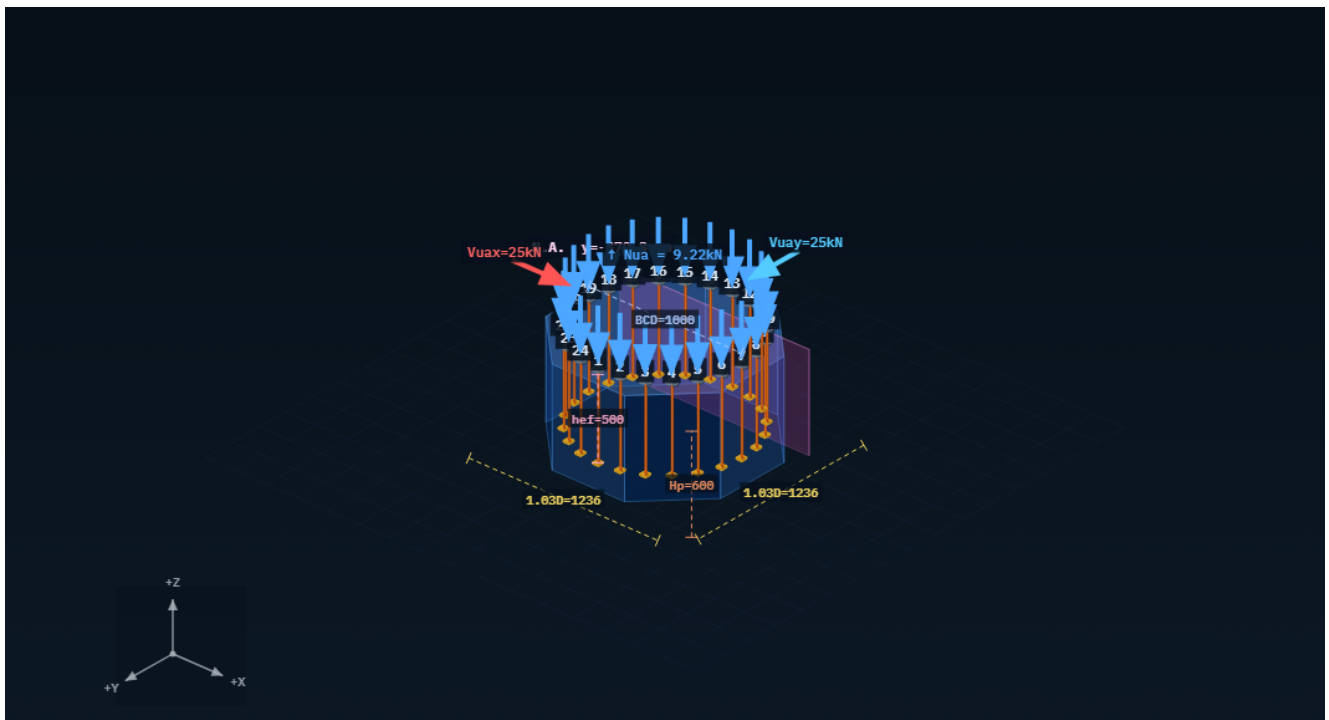


Anchor Bolt Structural Design

ACI 318-19 Chapter 17 — Anchoring to Concrete (v2.0, ANc precise formula)

Job No.: —
Project: —
Doc. No.: —
Item: —
Engineer: —
Date: 2026-08-03

3D Model View — Isometric View



Result Summary					
Failure Mode	Symbol	Capacity	Demand	Ratio	Status
T1 Steel (Tension)	ϕN_{sa}	145.05 kN	9.22 kN	6.40%	O.K
T2 Breakout (Tension)	ϕN_{cbg}	80.73 kN	92.39 kN	114.40%	N.G
T3 Pullout	ϕN_{pn}	148.34 kN	9.22 kN	6.20%	O.K
T4 Side - face	ϕN_{sb}	103.50 kN	92.39 kN	89.30%	O.K
V1 Steel (Shear)	ϕV_{sa}	93.12 kN	1.47 kN	1.60%	O.K
V2 Breakout (Shear)	ϕV_{cbg}	61.64 kN	1.47 kN	2.40%	O.K
V3 Pryout	ϕV_{cpg}	161.46 kN	1.47 kN	0.90%	O.K
Combined T + V	IR	1.14	1	114.40%	N.G

1. DESIGN INPUT CONDITIONS

Parameter	Symbol	Value / Description	Unit
Anchor Type	—	Cast - in Headed Bolt	
No. of Anchors (nx ny)	24 × 1	24.00 ea — GROUP	ea
Bolt Diameter	da	20.00	mm
Effective Embed. Depth	hef	500.00	mm
Neutral Axis Method	N.A.	Linear Stress Method	
Linear Stress Assumption	C / T	Concrete compression only + anchor tension only; c solved from C = T + P and k from moment equilibrium	
Modified hef	hef'	78.67 (Modified)	mm
hef' Calc.	—	= R s_X,max Group check: s_X,max < 3hef 130.5 < 1500.0 Group edge count (< 1.5hef) = 4 ca,max = 118.0 mm s_X,max = 130.5 mm hef' = max (ca,max / 1.5, s_X,max / 3) = max (78.7, 43.5) = 78.7 mm	
Head Width (AC)	eh	34.60	mm
Bearing Area	Abrg	883.0 = eh² - ϕ / 4 × da²	mm²
futa (limited)	futa	min (860, 1.9 × 400, 860) = 760 MPa	MPa
fya	fya	400.00	MPa
Steel Element Behavior	—	Ductile steel element governs	Table 17.5.3 (a)
Pedestal Lpx	Lpx	1236.00	mm
Pedestal Lpy	Lpy	1236.00	mm
Pedestal Height	Hp	600.00	mm
Concrete Strength	f'c	30.00	MPa
Lightweight Factor	λ	1.00	—
Crack Condition	—	Cracked	
Edge Distance X	Ca1,x	(1236 - 1000) / 2 = 118 mm	mm
Edge Distance Y	Ca1,y	(1236 - 1000) / 2 = 118 mm	mm
Governing Ca1	Ca1	min (118, 118) = 118 mm	mm

Design Tension	Nua	9.22	kN
Design Shear X	Vua,x	25.00	kN
Design Shear Y	Vua,y	25.00	kN
Resultant Shear	Vua	35.35 kN	kN
Rebar Dia.	dv	None	
Total Rebar	Ntot	0.00	ea
N1 (within 0.5hef)	N1	0.00	ea
0.5hef Zone	0.5hef	250.00	mm
Rebar fy	fy	400.00	MPa
Cover C1	C1	50.00	mm
Cover C2	C2	75.00	mm
Tie Bar	dtie	D16	
ϕ Steel Tension	ϕs,t	0.75 (ductile)	Table 17.5.3 (a)
ϕ Steel Shear	ϕs,v	0.65 (ductile)	Table 17.5.3 (a)
ϕ Conc. Tension	ϕc,t	0.70	Table 17.5.3 (b)
ϕ Conc. Shear	ϕc,v	0.70	Table 17.5.3 (b)

2. TENSILE STRENGTH — ACI 318-19 §17.6

T1. Steel Failure in Tension [ϕNsa] — §17.7.1.2

Item	Sym.	Formula / Calculation	Value	Ref.
Bolt dia. 0.9da	0.9da	$0.9 \times 20 = 18.0 \text{ mm}$	18.00 mm	§17.6.1.2
Eff. tensile area Ase,N	Ase,N	$\phi / 4 \times (0.9da)^2 = \phi / 4 \times 18.0^2$	254.47 mm²	§17.6.1.2
Limited futa	futa	$\min (860, 1.9 \times 400, 860) = 760 \text{ MPa}$	760.00 MPa	§17.6.1.2
Nom. strength (1ea)	Nsa,1	$Ase,N \times futa = 254.47 \times 760 / 1000$	193.40 kN	§17.7.1.2
ϕ steel tension	ϕs,t	Ductile steel element governs	0.75	Table 17.5.3 (a)
ϕNsa (governing)	ϕNsa	$\phi \times n \times Ase,N \times futa = 0.75 \times 24 \times 254 \times 760 / 1000$	145.05 kN	§17.7.1.2

[O.K] ϕNsa Nua: 145.05 kN 9.22 kN (6.4%)

T2. Concrete Breakout in Tension [φNcbg] — §17.6.2.1b				
Item	Sym.	Formula / Calculation	Value	Ref.
Projected Area ANc — Circular anchor group annular area (v2.0 fix)				
ANc0	ANc0	$9 \times hef^2 = 9 \times 78.7^2$	55696.00 mm²	§17.6.2.1.4
Outer diameter	Do	$BCD + 2Ca1 = 1000 + 2 \times 118$	1236.00 mm	PIP STE05121 Fig.B - 1
Inner diameter	Di	$\max (0, BCD - 3hef) = \max (0, 1000 - 3 \times 79)$	764.00 mm	PIP STE05121 Fig.B - 1
ANc (raw, annular)	ANc,raw	$\phi / 4 \times (Do^2 - Di^2) / 2$	370708.00 mm²	Annular area
ANc (capped)	ANc	$\min (ANc,raw, n \times ANc0) = \min (370708, 1336704)$	370708.00 mm²	§17.6.2.1.2
ANc / ANc0	ratio	370708 / 55696	6.66	
Basic Concrete Breakout Strength Nb				
kc	kc	cast - in = 10 / post - installed = 7	10.00	§17.6.2.2.2
Lambda	λ	Lightweight factor	1.00	§19.2.4
f'c	f'c	30	5.48	
hef^1.5	hef^1.5	$78.7^{1.5}$	697.73	
Nb	Nb	$kc \times \lambda a \times f'c \times hef^{1.5} = 10 \times 1 \times 5.4772 \times 697.73 / 1000$	38.22 kN	§17.6.2.2.1
Modification Factors				
ψec,N	ψec,N	No eccentricity 1.0	1.00	§17.6.2.3
ψed,N	ψed,N	$Ca1 \ 1.5hef \ 1.0$	1.00	§17.6.2.4
ψc,N	ψc,N	Cracked 1.0	1.00	§17.6.2.5
ψcp,N	ψcp,N	Cast - in 1.0	1.00	§17.6.2.6
Breakout Strength				
Ncbg	Ncbg	$(ANc / ANc0) \times \psi ec \times \psi ed \times \psi c \times \psi cp \times Nb$	115.33 kN	§17.6.2.1b
φc,t	φc,t	Supplementary Reinforcement: No	0.70	Table 17.5.3 (b)
φNcbg	φNcbg	0.7×115.3312	80.73 kN	§17.6.2.1b
[N.G] φNcbg Nua: 80.73 kN < 9.22 kN (114.4%)				

T3. Pullout Failure [φNpn] — §17.6.3.1				
Item	Sym.	Formula / Calculation	Value	Ref.
Bearing Area	Abrg	$eh^2 - \phi / 4 \times da^2 = 34.6^2 - \phi / 4 \times 20^2$	883.00 mm²	§17.6.3.2.2
Np (1 anchor)	Np	$8 \times Abrg \times f'c = 8 \times 883 \times 30 / 1000$	211.92 kN	§17.6.3.2.2
ψc,P	ψc,P	Cracked 1.0	1.00	§17.6.3.3
φNpn	φNpn	$\phi \times \psi_{c,P} \times Np \times n = 0.7 \times 1 \times 211.9200 \times 24$	148.34 kN	§17.6.3.1
[O.K] φNpn Nua: 148.34 kN 9.22 kN (6.2%)				
T4. Side-face Blowout [φNsb] — §17.6.4.1				
Item	Sym.	Formula / Calculation	Value	Ref.
Applicability	—	Ca1 < 0.4hef 118 < 200 mm	APPLIES	§17.6.4.1
Nsb	Nsb	$160 \times Ca1 \times Abrg \times \lambda \times f'c$	147.85 kN	§17.6.4.1
φNsb	φNsb	0.7×147.8495	103.50 kN	§17.5.3
[O.K] φNsb Nua: 103.49 kN 9.22 kN (8.9%)				
3. SHEAR STRENGTH — ACI 318-19 §17.7				
Resultant Shear Force				
Item	Sym.	Formula	Value	Ref.
Shear X	Vua,x	Input	25.00 kN	input
Shear Y	Vua,y	Input	25.00 kN	input
Resultant	Vua	$(Vuax^2 + Vuay^2) = (25^2 + 25^2)$	35.35 kN	§17.7
V1. Steel Failure in Shear [φVsa] — §17.7.1.2b				
Item	Sym.	Formula / Calculation	Value	Ref.
Eff. shear area Ase,V	Ase,V	$\phi / 4 \times da^2 = \phi / 4 \times 20^2$	314.16 mm²	§17.7.1.2
futa (limited)	futa	760	760.00 MPa	§17.6.1.2
φ steel shear	φs,v	Ductile steel element governs	0.65	Table 17.5.3 (a)
φVsa	φVsa	$\phi \times 0.6 \times Ase,V \times futa \times n = 0.65 \times 0.6 \times 314 \times 760 \times 24 / 1000$	93.12 kN	§17.7.1.2b
[O.K] φVsa Vua: 93.12 kN 35.36 kN (1.6%)				

V2. Concrete Breakout in Shear [φVcbg] — §17.7.2.1b				
Item	Sym.	Formula / Calculation	Value	Ref.
Projected Area AVc				
AVc0	AVc0	$4.5 \times Ca1^2 = 4.5 \times 118^2$	62658.00 mm²	§17.7.2.1.3
Height hc	hc	$\min (Hp, 1.5Ca1) = \min (600, 177)$	177.00 mm	§17.7.2.1.2
AVc	AVc	$AVc = \min (1.5ca1D, n \times 4.5ca1^2) = \min (212400.0, 1503792.0) = 212400.0 \text{ mm}^2$	212400.00 mm²	PIP STE05121 Fig.B - 1
AVc / AVc0	ratio	212400 / 62658	3.39	
Basic Shear Strength Vb				
le	le	$\min (8 \times da, hef) = \min (160, 79)$	160.00 mm	§17.7.2.2.1
Vb1	Vb1	$0.60 \times (le / da)^{0.2} \times da \times \lambda \times fc \times Ca1^{1.5}$	28.55 kN	§17.7.2.2.1
Vb2	Vb2	$3.7 \times \lambda \times fc \times Ca1^{1.5}$	25.98 kN	§17.7.2.2.1
Vb	Vb	$\min (Vb1, Vb2)$	25.98 kN	—
Modification Factors				
ψec,V	ψec,V	No eccentricity	1.00	§17.7.2.3
ψed,V	ψed,V	$Ca2 \ 1.5Ca1' \ 1.0$	1.00	§17.7.2.4
ψc,V	ψc,V	Cracked 1.0	1.00	§17.7.2.6
ψh,V	ψh,V	$Hp \ 1.5Ca1' \ 1.0$	1.00	§17.7.2.5
Breakout Shear Strength				
Vcbg	Vcbg	$(AVc / AVc0) \times \psi_{ec} \times \psi_{ed} \times \psi_c \times \psi_h \times Vb$	88.06 kN	§17.7.2.1b
φc,v	φc,v	Supplementary Reinforcement: No	0.70	Table 17.5.3 (b)
φVcbg	φVcbg	0.7×88.0569	61.64 kN	§17.7.2.1b
[O.K] φVcbg Vua: 61.64 kN 35.36 kN (57.4%)				
V3. Pryout [φVcpg] — §17.7.3.1				
Item	Sym.	Formula / Calculation	Value	Ref.
kcp	kcp	hef 65mm 2	2.00	§17.7.3.1
Ncbg	Ncbg	(from T2, v2.0 annular ANc)	115.33 kN	T2

Vcpg	Vcpg	$kcp \times Ncbg = 2 \times 115.3312$	230.66 kN	\$17.7.3.1
$\phi Vcpg$	$\phi Vcpg$	0.7×230.6624	161.46 kN	\$17.5.3
[O.K] $\phi Vcpg$ Vua : 161.46 kN 35.36 kN (21.9%)				

4. COMBINED TENSION + SHEAR — ACI 318-19 §17.8

Item	Sym.	Formula / Calculation	Value	Ref.
ϕNn (governing)	ϕNn	$\min (\phi Nsa, \phi Ncbg, \phi Npn, \phi Nsb)$	80.73 kN	\$17.8.1
ϕVn (governing)	ϕVn	$\min (\phi Vsa, \phi Vcbg, \phi Vcpg)$	73.22 kN	\$17.8.1
Tension Ratio	$Nua / \phi Nn$	$9.221\sim 92.393 / 80.732$	1.14	\$17.8.3
Shear Ratio	$Vua / \phi Vn$	$1.473\sim 35.355 / 73.222$	0.02	\$17.8.3
Interaction Mode	—	—		\$17.8.3
Interaction Ratio	IR	Single dominant	1.14 1	\$17.8.3
[N.G] IR 1: 1.1444 > 1 (114.4%)				